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Anthropometry of Brazilian Air Force pilots

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ABSTRACT

Anthropometric data are essential for the design of military equipment including sizing of aircraft cockpits and personal gear. Currently, there are no anthropometric databases specific to Brazilian military personnel. The aim of this study was to create a Brazilian anthropometric database of Air Force pilots. The methods, protocols, descriptions, definitions, landmarks, tools and measurements procedures followed the instructions outlined in *Measurer's Handbook: US Army and Marine Corps Anthropometric Surveys, 2010–2011* – NATICK/TR-11/017. The participants were measured countrywide, in all five Brazilian Geographical Regions. Thirty-nine anthropometric measurements related to cockpit design were selected. The results of 2133 males and 206 females aged 16–52 years constitute a set of basic data for cockpit design, space arrangement issues and adjustments, protective gear and equipment design, as well as for digital human modelling. Another important implication is that this study can be considered a starting point for reducing gender bias in women's career as pilots.

Practitioner Summary: This paper describes the first large-scale anthropometric survey of the Brazilian Air Force pilots and the development of the related database. This study provides critical data for improving aircraft cockpit design for ergonomics and comprehensive pilot accommodation, protective gear and uniform design, as well as digital human modelling.

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Anthropometry; cockpit design; flight deck; pilot accommodation; Brazilian Air Force

1. Introduction

Anthropometry can be used to help ensure fit for a broad range of target populations by providing the range and limits of the target population in diverse markets (Hsiao 2013). Many everyday consumer products including clothes, office workspaces, vehicles and assembly lines need to be adjusted to user anthropometrics to maximise the usability and minimise the negative effects on the user (Chuan, Hartono, and Kumar 2010; Lars et al. 2009; Lovesey 2006; Singh et al. 1995).

Anthropometric data are used to provide body size information for military and civilian population research and equipment development. Knowledge of body size and proportions is essential for the design and sizing of military uniforms, protective, and personal equipment. Anthropometric data are also critical for engineering human factors in military equipment including aircraft, vehicles, tanks, submarines and many other types of weapons systems (Department of Defense 1991). Thus, anthropometric measurement of the military population provides a standard, statistical description of that population and has been used to define the accommodation envelope,

compatibility, integration and safety of military equipment as well as commercial products, and ultimately to improve the performance and logistic efficiency in equipment systems that involve a human component (Department of Defense 1991; Zhuang and Bradtmiller 2005).

A critical, but often overlooked, step in the ergonomic design process is determining the intended users of the equipment, or defining the target population (HFES 300 Committee 2004; Zhuang and Bradtmiller 2005; Zhuang, Bradtmiller, and Shaffer 2007). Appropriate accommodation of body size in aircraft cockpits is still being sought in military services despite many years of experience by designers (Kennedy and Zehner 1995). Aircraft cockpits are one of the most complex workstations seen today. However, the bulk of military aircraft in use worldwide are built around the anthropometry of western aviators, predominantly from the US (Singh et al. 1995). Currently, the Brazilian Air Force and Brazilian Aeronautical Industry are designing military aircraft using US military anthropometric data (da Silva 2013). According to Kennedy (1982), non-accommodation in an aircraft not only produces conditions under which the product is inconvenient to

operate, but also in which the user's safety and the basic mission of the aircraft can be compromised.

The following six dimensions have been considered critical to cockpit design for aircraft ejection seats in several studies, and are also currently used to evaluate cockpit accommodation and for pilot selection in many Air Forces worldwide: sitting height; eye height, sitting; shoulder height, sitting; knee height, sitting; buttock-knee length; and functional reaching (Department of Defense 2008; Hudson, Zehner, and Meindl 1998; Kennedy and Zehner 1995; Roebuck 1995; Zehner 1996; Zehner, Meindl, and Hudson 1993). These body dimensions are related to provide appropriate internal and external field of view, to maintain a safe clearance of the cabin structures, mainly in case of ejection, and to reach all the controls. Since, in aviation most controls request grasping and handling precision, the use of the thumb is crucial as stated by Melmoth and Grant (2012).

The ergonomic design process requires that we know the body dimensions relevant to the function of the equipment as well as whom it should fit. However, these body dimensions are only helpful if they are taken from a sample population that represents the body size variation expected in the target population (HFES 300 Committee 2004). In this context, it is important to point out that there are no published anthropometric data collected from Brazilian military personnel, including the Air Force. Furthermore, only two anthropometric databases from any Brazilian populations were found in the literature. The first database dates from 1988, when measurements were taken of male industrial workers from Rio de Janeiro; however, only a few dimensions were physically measured, while others were mathematically estimated (Ministry of Science and Technology 1988; Pheasant and Haslegrave 2006). Another database from 2008 was found at the Brazilian Civil Aviation Agency (ANAC) web page (Silva and Monteiro 2009), describing eight anthropometric dimensions taken from Aviation Transport Users (common passengers).

HFES 300 Committee (2004) states that considering the need to incorporate body size and proportional information in a design, it is always preferable (when affordable) to create a detailed anthropometric database, comprising accurate measurement of relevant dimensions, on a sample population encompassing the breadth of the intended user population (in this case, male and female pilots). Thus, a critical step in achieving improved design and military function is the measurement of the anthropometric data across this population, and the development of a National Anthropometric Database of Brazilian Pilots.

As mentioned by Gordon et al. (2013), anthropometric data for design are most useful when they are representative of the user population for which the designs will be created (Gordon et al. 2013; Lars et al. 2009; Lee et al. 2013).

Therefore, the lack of current, accurate data describing the body dimensions of its aircrew prompted the Brazilian Air Force Command Staff (EMAER) to initiate a large-scale anthropometric survey of aircrew. In May 2013, the EMAER issued an Ordinance (number 17/3SC, 06/05/2013)² formalising the survey and providing knowledge to all military organisations in the Brazilian Air Force (in Portuguese, Força Aérea Brasileira – FAB) (Ministry of Defense 2013). At the beginning of the plan, it was intended to have a census covering 100% (3350 subjects) of the FAB aircrew, but it was not feasible due to the complexities of general military duties, as well as military operations. A sample of the target population needed to be constructed.

This study aimed to create a Brazilian anthropometric database of FAB pilots for improving aircraft cockpit design for ergonomics and comprehensive pilot accommodation, protective gear and uniform design, as well as digital human modelling.

2. Methods

2.1. Subjects

The sample consisted of 2339 FAB aircrew (2133 males and 206 females) aged 16–52 years. The target population for this study was the Brazilian Air Force Pilot Population, comprised of two subpopulations: the 'Potential Pilot Population' (847 males and 188 females) and the 'Current Brazilian Air Force Pilots' (1286 males and 18 females).

The aim of including the 'Potential Pilot Population' was to capture a more diverse sample of body sizes and proportions as well as a more comprehensive age range. This population included: (a) current cadets (pilot trainee) in the Air Force Academy – AFA (including both genders), (b) students from the Preparatory School of Aviator Cadets (EPCAR) in their third and final high school year before going to AFA, and (c) female administrative officers. The final group was included in order to increase the number of females in the study, since currently there are only 28 female pilots in the FAB, as well as to capture diverse body sizes and proportions.

The 'Current Brazilian Air Force's (FAB) pilots' population considered pilots graduated from 1984 to 2012 encompassing ranks from Lieutenant to Colonel.

Considering that demographic effects on anthropometry have been reproducibly reported in the literature to be associated with differences in body dimensions (Hotzman et al. 2011; Hsiao, Long, and Snyder 2002), some demographic information were acquired from each subject, not only for this study, but also for further investigations, which included: military service component, military rank, age, racial/ethnic background, parents' racial/ethnic background and profession, years of military service, aircraft qualifications, and gender. Due to the broad

miscegenation, the large European and Asian immigration, and the huge variability in socio-economic conditions in Brazil, it is necessary to know the parents' racial/ethnic background since some specific racial features could mislead the data interpretation (e.g. white person with short torso and long limbs). The parents' profession is a way to figure out the social level the participant came from, since the socio-economic situation can be associated with quantity and quality of dietary intake during childhood and this maybe could influence growth and anthropometric data (Gross et al. 1990). This specific demographic feature will be further investigated in a future study.

The participants were measured countrywide, in all five Brazilian Geographical Regions (North, Northeast, South, Southeast and Central-west), at 80 Administrative Military Organizations, two Formation Schools, 14 Air Force Bases and 35 Squadrons, thus the sample included female administrative officers, cadet pilots and non-pilots, and FAB pilots, including instructor, fighter, helicopter, patrol, recognition and transport pilots.

2.2. Subject acquisition/selection process

Power analyses for both men and women were conducted to ensure sufficient sample sizes to accurately describe design requirements for each sex whenever appropriate.

The minimum sample size was determined using the procedures (Equations 1 and 2) outlined in International Organization for Standardization (ISO) 15535:2012 – General Requirements for Establishing Anthropometric Databases (ISO 2012) and Natick TR-13/018 – 2010 Anthropometric Survey Of US Marine Corps Personnel: Methods and Summary Statistics (Gordon et al. 2013).

For most engineering databases, the most challenging parameter estimates are the 5th and 95th percentiles, which are commonly used to establish accommodation boundaries in military design and sizing requirements (Gordon et al. 2013).

Equation (1) is a simplified expression of the minimum sample size required for estimation of 5th–95th percentiles with 95% confidence and a precision of 1% of the mean value. Equation (2) gives the minimum sample sizes needed to estimate mean values with 95% confidence and 1% precision (Gordon et al. 2013; ISO 2012). Therefore, Equation (1) is essential for determining the minimum sample size for the purpose of this study, rather than Equation (2).

$$n \geq (1.96 * CV)^2 * (1.534)^2 \quad (1)$$

$$n \geq (1.96 * CV)^2 \quad (2)$$

where n = minimum sample size, and CV = Coefficient of Variation = (Standard Deviation/Mean) * 100.

A worst case approach is taken by choosing those critical body dimensions with the highest CVs to establish minimum sample sizes, knowing that the results will be more than sufficient for less variable body dimensions. As stated by Gordon et al. (2013), previous military surveys have shown that of the most commonly used body dimensions, the two with the largest CVs (therefore requiring the largest sample sizes for precise parameter estimates) are Waist Circumference and Chest Circumference. Since the main objective of the study concerns cockpit design, the six critical dimensions for this purpose were also used for minimum sample size calculation.

The sampled male population was well above the calculated requirement using both equations for all dimensions. For the female population, the requisite minimum sample size also has been fulfilled for the six critical dimensions using both equations, but in the 'worst scenario' – waist circumference and chest circumference – the required minimum sample size was larger than the entire female FAB population ($n = 806$ and 602 , respectively). Thus, even a 100% participation rate from the female population ($n = 319$) would not be enough for accurate estimation of 5th–95th percentiles with 95% confidence and 1% precision, using the aforementioned equations.

However, it can be inferred that for the purpose of this study the achieved female sample size was sufficient for estimating mean and 5th–95th, considering that the sample acquired has demographic characteristics similar to the entire population.

2.3. Relevant body dimensions

A total of thirty-nine (39) critical anthropometric dimensions (using traditional measures of the body) were selected for the database (see Figure 1). Definitions used were those published in Natick TR-11/017 – Measurer's Handbook: US Army and Marine Corps Anthropometric Surveys, 2010–2011. These dimensions were selected since they relate to cockpit design, protective gear design and digital human modelling, in accordance with the literature review (Department of Defense 2008; Hudson, Zehner, and Meindl 1998; Kennedy and Zehner 1995; Lovesey 2006; Roebuck 1995; Zehner 1996, 2002; Zehner, Meindl, and Hudson 1993).

(1) Abdominal extension depth, sitting; (2) Acromial height, sitting; (3) Biacromial breadth; (4) Biceps circumference, flexed; (5) Bideltoid breadth; (6) Buttock circumference; (7) Buttock – knee length; (8) Buttock – popliteal length; (9) Calf circumference; (10) Cervical height; (11) Chest circumference; (12) Chest depth; (13) Elbow rest height; (14) Eye height, sitting; (15) Foot breadth; (16) Foot length; (17) Forearm circumference, flexed; (18) Forearm – forearm breadth; (19) Forearm – hand length; (20) Functional leg length; (21) Hand Breadth;

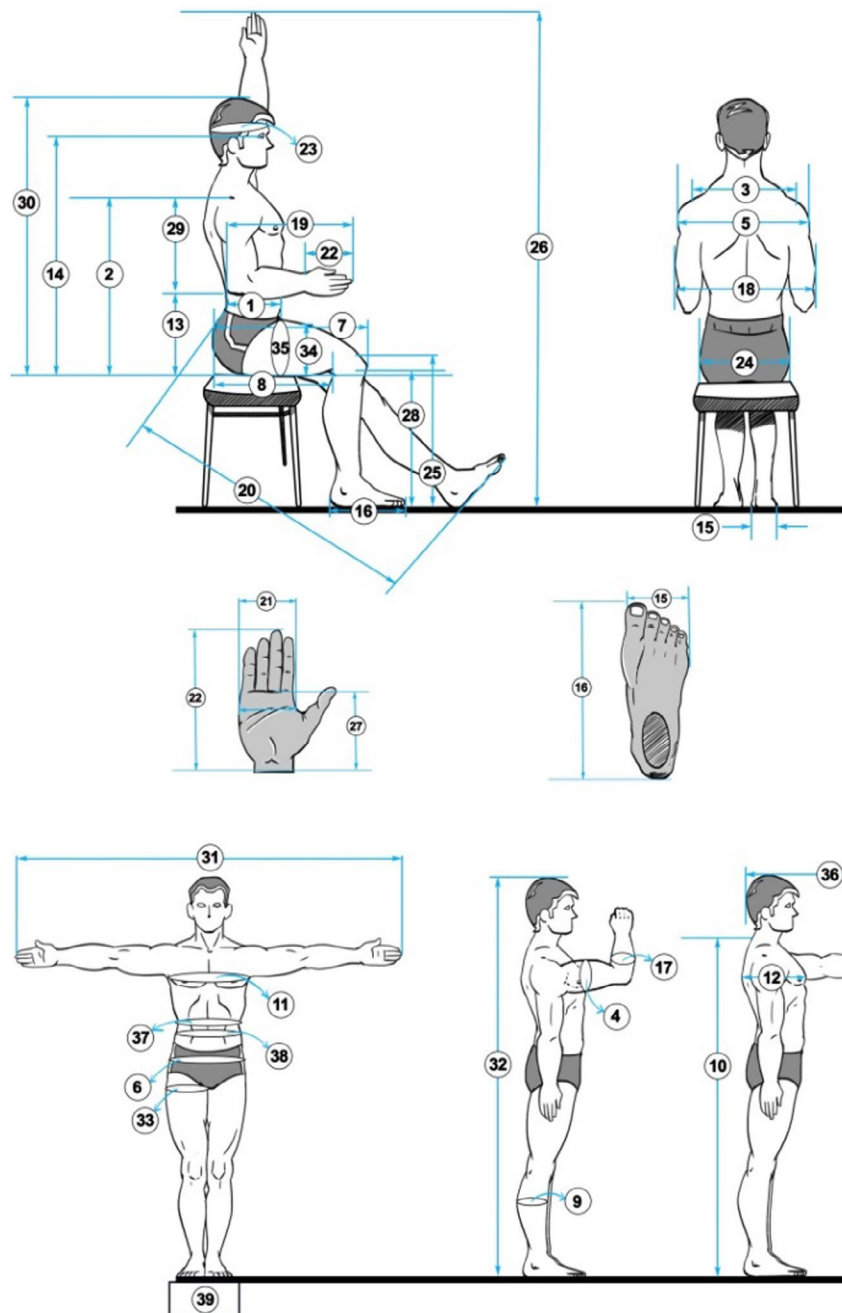


Figure 1. Measured dimensions.

(22) Hand length; (23) Head circumference; (24) Hip breadth, sitting; (25) Knee height, sitting; (26) Overhead fingertip reach; (27) Palm length; (28) Popliteal height, sitting; (29) Shoulder – elbow length; (30) Sitting height; (31) Span; (32) Stature; (33) Thigh circumference; (34) Thigh clearance; (35) Thigh max circumference, sitting; (36) Thumbtip reach; (37) Waist circumference (narrower); (38) Waist circumference (Omphalion); (39) Weight.

2.4. Measurement procedures

The measuring team consisted of personnel currently responsible for the anthropometric measurements

standards procedures used in the annual FAB fitness test, very familiar with the landmarks and general procedures used for anthropometric data collection. They were trained over a period of two weeks to familiarise them with the specific measuring technique and the equipment.

In order to allow further comparisons, the methods, protocols, descriptions, definitions, landmarks, tools and measurements procedures followed the instructions outlined in Natick TR-11/017 – Measurer's Handbook, which is also used by the US Air Force (Hotzman et al. 2011). These standard procedures are useful for comparing data-sets between nations, and are measured according to the protocol in International Organization for Standardization

(ISO) 7250-1:2008 – Basic human body measurements for technological design-Part 1-Body measurement definitions and landmarks and ISO 15535:2012 (Hotzman et al. 2011; ISO 2008, 2012).

Two months before, all FAB organisations were requested to access a URL on the Brazilian Air Force Staff webpage, linking to information on the Anthropometric Survey. In addition, all of them were notified in advance, whereupon they received the survey schedule and descriptions of the study. On the day before measurement, participants were briefed (a 30 min PowerPoint™ presentation) on the general purpose and procedural aspects of the survey, and filled out the biographic/demographic data form. All questionnaires were coded in order to protect the data and avoid subject identification.

There were four distinct measuring stations and two measurers per station (alternating as measurer and recorder), thus each team consisted, on average, of 12 people, with one 'substitute measurer' per station. The four stations were divided as following: (1) Landmarking; (2) Standing and sitting measurements (depths, breadths and lengths) with the Beam caliper – including foot measurements with the Modified Brannock device; (3) Standing and sitting measurements (heights and lengths) with the anthropometer; and (4) Circumferences and hand measurement with the Poech sliding caliper. Measurements in sitting position were taken using a 70 cm height cushionless table, two wood boxes as footrest (with 40 and 35 cm height), styrofoam pieces of various thicknesses to move the feet up and down, and a wood buttock plate. Where there was a choice of right or left, all measurements were taken on the right side unless otherwise specified or in the rare cases where an injury or anatomical abnormality made it necessary to measure on the left side, in this case a comment was recorded indicating that the dimension was measured on the left side. All measurements were made to the nearest millimetre. Weight was taken to the nearest 0.1 kg.

Before leaving each measuring station, participants had their datasheet double checked for missing values in order to ensure complete data collection. In each measuring station there was a banner with pictures showing the dimensions to be measured and the right sequence to be performed.

At each station there was a laptop for digital data entry, and the written data sheet was kept as back-up. In order to double check unusual values (outliers), a filtering process for each dimension was created, in Excel™, based on databases published on ISO 7250-2:2010 and DOD-HDBK 743A (Department of Defense 1991; ISO 2010). Values were verified before the subject left the station and if the filter identified any atypical values the subject was re-measured.

Males were measured wearing swimsuits, bare-chested. Females were measured wearing tight sports-shorts (like

bicycle shorts) and a sports-bra (halter top). All subjects were measured between 24 July 2013 and 30 July 2014.

2.5. Data quality

In order to enable the measurement teams to acquire the measurements in the same way, and to minimise inter-measurer error, an extensive and comprehensive practical training was undertaken until all the measurers successfully learned the tasks. Prior to conducting the measurer training, the Measurer's Handbook from Hotzman et al. (2011) was translated into Portuguese. The first week was online, allowing participants to familiarise themselves with the Natick manual and second week training was hands on, with about 30 h of practical training, until the criteria for the Natick allowable observer error was reached, in order to ensure data reproducibility.

In the interest of ensuring the quality of the collected data an expert (one physical therapist with experience in anthropometric measurement using the Natick protocol) was used as Criterion Measurer in order to serve as a cross-validation method. The Criterion Measurer is an experienced measurer who serves as the standard, in order to assess the performance of measurers in training and during the survey, against the allowable observer error defined in Appendix G of the Natick Measurer's Handbook (Hotzman et al. 2011).

The quality of measurements of each measurer was gauged by ensuring that their measuring values were kept as equal to or below the allowable observer error, when compared to the unique Criterion Measurer. Therefore, it can be accepted that the data presented in the results are comparable values, regardless of the personal measuring style of each individual on the team.

The allowable observer errors were used for three purposes: during the initial training period, as an indicator that measurers had successfully learned their tasks; to 'recalibrate' the team at the beginning of each new survey location; and as a standard for daily error checks (the measuring error could not exceed the allowable observer error for any given dimension).

Thus, the Criterion Measurer and the allowable observer error values established the minimum reliability criteria for the data collected in the survey.

During each measuring day, some randomly chosen measurements were repeated for some subjects to track and manage observer error. Each day, a randomly chosen subject underwent measurement a second time at any given station, meaning that three subjects per day underwent additional measurement (one for each measuring station). If the measurement was larger than the allowable error, then a third measurement had been taken. Considering a total of 30–40 participants measured per

Table 1. Sample distribution ($n = 2339$).

Participants	Males		Females		Total	
	Frequency	*Percent	Frequency	*Percent	Frequency	*Percent
Aviation officers ^a	1286	59.0	18	64.3	1304	59.0
Administrative officers ^b	00	00	104	50.3	104	50.3
Cadets ^b	643	100	84	100	727	100
EPCAR students ^b	204	98.6	00	00	204	98.6
Total	2133	70.4	206	64.6	2339	69.8

a = Current Brazilian Air Force Pilots Population ($n = 2209$).

b = Potential Pilot Population ($n = 1141$).

*percentage of the whole Air Force Pilots Population (a and b). EPCAR = Preparatory School of Aviator Cadets.

Table 2. Age distribution of participants compared to whole FAB population.

Age group	Participants			
	Male		Female	
	Sample ($n = 2133$)	Population ($n = 3031$)	Sample ($n = 206$)	Population ($n = 319$)
	Percent	Percent	Percent	Percent
<21	18.3	17.9	11.2	13.2
21–25	26.6	19.6	40.8	33.2
26–30	16.1	17.1	24.8	29.2
31–35(38)*	15.2	14.7	23.2	24.4
36–40	7.8	11.1	0	0
41–45	11.1	10.7	0	0
>45	4.9	8.9	0	0
Total	100	100	100	100
Minimum age	16		18	
Maximum age	52		38	

*38 years-old is the highest age among females in the FAB.

day, approximately 10% of the participants were double checked.

2.6. Anthropometric instrument

A set of traditional anthropometry measurement devices was used to take 39 traditional body measurements. These devices were an anthropometer (GPM™, Switzerland), a beam caliper (rearranged pieces of the anthropometer), Poech sliding caliper (GPM™, Switzerland), a Lufkin™ Executive Diameter metric tape measure-W606PM (Cooper Tools, Apex, NC), a Filizola™ scale PL 200, digital, maximum weight of 200 kg, and 0.1 kg precision, a Modified Brannock device, and a Wall chart. The Modified Brannock device for measuring the foot more easily and precisely, and the wall chart (sheet of laminated millimetre graph paper) used for obtaining measures of functional arm reach and span.

2.7. Preparing data and statistics

Prior to the creation of summary statistics tables, all raw data were reviewed and scrutinised for unusual values, following the procedures outlined in International Organization for Standardization (ISO) 7250-2 – Basic

human body measurements for technological design-Part 2-Statistical summaries of body measurements (ISO 2010). The minima and maxima for each dimension were re-examined, comparing them with the minima and maxima from other databases, helping to verify the existence of any erroneous measurement values and to find any outlying values.

In order to find outliers, in accordance with Annex F of the ISO 15535 (2012), the mean value and the standard deviation of each age group were used to individually review the measurements, to identify data that fell outside of the mean $\pm 3SD$. Measurements were plotted in scatterplots and boxplots, in pairs, by age group, and visually inspected for anomalies. Subjects identified by these definitions as outliers were further investigated. In case of any discrepancy, the cause was evaluated and data were corrected if the cause was identifiable, and dropped if not. If the dropped measurement corresponded to one of the six cockpit critical dimensions, the entire subject was dropped.

After review and scrutiny of the raw data, correcting and dropping data points as necessary following the procedures described, the resulting data formed the reference data-set, which was used for the database creation, statistical analysis and testing.

3. Results

Initially, a census had been planned to measure 100% of the FAB aircrew (3350 subjects – 3031 males and 319 females). However, due to the complexities of general military duties, as well as military operations, 2133 out of 3031 (70%) males and 206 out of 319 (65%) females represent the measured sample (participants). Table 1 shows the distribution of the sample with respect to occupation and gender.

Tables 2 and 3 summarise the characteristics of the FAB anthropometric survey participants, outlining age, gender and ethnicity. The age distribution of FAB survey participants ranged from 16 to 52 for males, and from 18 to 38 for females. The age ranges for males and females are quite different, as could be seen in Table 2. It is noteworthy

Table 3. Racial/ethnic distribution of FAB survey participants.

Race/ ethnicity	Males		Females		Total	
	Frequency	Percent	Frequency	Percent	Frequency	Percent
White	1548	72.6	137	66.5	1685	72.0
Brown	466	21.8	55	26.7	521	22.3
Black	93	4.4	9	4.4	102	4.4
Asian	20	0.9	4	1.9	24	1.0
Indigenous	4	0.2	1	0.5	5	0.22
Others	2	0.1	0	0	2	0.08
Total	2133	100	206	100	2339	100

that females are recent in the FAB history (Administrative Officers since 1996 and Pilots since 2004).

Table 2 shows both age distributions of FAB population and FAB Survey participants (Sample) for comparison purpose.

Comparing the age distribution of FAB survey participants with the FAB population (Table 2) shows a very similar distribution except for an over-representation of the 21–25 male age group and an under-representation

Table 6. Subjects' constraints of FAB survey non-participation.

Constraints	Male non-participants	Female non-participants
	Frequency	Frequency
Strategy and Politics Course (Colonel)	58	0
Air Staff and Command Course (Major)	80	0
Maneuver Captains Career Course (Captain)	43	0
Flight missions	313	5
Vacations	83	21
Abroad missions	108	0
Military organisations not visited	166	31
*Others	46	48
Pregnancy	0	8
Total	897	113

*Mourning, paternity, sickness, safety service schedule, military operations, otherE courses.

of the 36–40 and above 45 male age groups. This can be explained by the higher participation of the cadets (100%) and by the lower relative participation of about 80 majors,

Table 4. FAB survey participants by branch.

Rank	Branch	Males		Females		Total	
		Frequency	Percentage	Frequency	Percentage	Frequency	Percentage
Officers	Fighter	151	7.08	02	0.97	153	6.54
	Transport	329	15.43	10	4.85	339	14.49
	Helicopter	128	6.00	05	2.43	133	5.69
	Instructor	95	4.45	01	0.49	96	4.11
	Rec-tact ^a	41	1.92	00	00	41	1.75
	Inspector ^b	45	2.11	01	0.49	46	1.97
	SAR ^c	23	1.08	00	00	23	0.98
	Patrol	23	1.08	00	00	23	0.98
	N.O.func ^d	451	21.14	00	00	451	19.28
	Administ ^e	00	00	103	50.00	103	4.40
Cadets and students	<i>Subtotal</i>	1286	60.29	122	59.22	1408	60.20
	Trainees ^f	461	21.61	18	8.74	479	20.48
	Administ ^e	101	4.74	66	32.04	167	7.14
	Infantry	81	3.80	00	00	81	3.46
	Students ^g	204	9.56	00	00	204	8.72
	<i>Subtotal</i>	847	39.71	84	40.78	931	39.80
Total		2133	100	206	100	2339	100

a. Reconnaissance; b. Inspection of the flight aids systems; c. Search and Rescue; d. Pilots in No Operational functions; e. Administrative; f. Pilot trainees; g. Last year of the FAB High School and future pilot trainees (Cadets).

Table 5. Military ranks of FAB survey participants.

Military rank	Males		Females		Total	
	Frequency	Percentage	Frequency	Percentage	Frequency	Percentage
Student (third year)	204	9.56	00	00	204	8.72
First year cadet	177	8.30	09	4.37	186	7.95
Second year cadet	158	7.41	23	11.17	181	7.74
Third year cadet	164	7.69	26	12.62	190	8.12
Fourth year cadet	144	6.75	26	12.62	170	7.27
Second Lieutenant	156	7.31	26	12.62	182	7.78
First Lieutenant	310	14.53	54	26.21	364	15.56
Captain	326	15.29	33	16.02	359	15.35
Major	176	8.25	09	4.37	185	7.91
Lieutenant Colonel	200	9.38	00	00	200	8.55
Colonel	118	5.53	00	00	118	5.05
Total	2133	100	206	100	2339	100

Table 7. Height and weight comparison of male and female participants with non-participants.

	Age group	Participants			Non-participants			Comparison	
		<i>n</i>	Weight mean	Stature mean	<i>n</i>	Weight mean	Stature mean	Delta weight	Delta stature
Male	<21	391	72	1748	6	70	1754	2	-6
	21-25	567	76	1752	192	77	1755	-1	-3
	26-30	344	82	1758	224	81	1758	1	0
	31-35/38*	325	83	1762	175	82	1761	1	1
	36-40	165	82	1755	105	81	1760	1	-5
	41-45	236	84	1757	103	82	1762	2	-5
	>45	105	85	1762	92	83	1766	2	-4
	Total	2133			897				
Female	<21	23	62	1640	0	-	-	-	-
	21-25	84	61	1658	29	61	1659	0	-1
	26-30	51	63	1652	54	63	1651	0	1
	31-35/38*	48	62	1648	30	62	1642	0	6
	36-40	-	-	-	-	-	-	-	-
	41-45	-	-	-	-	-	-	-	-
	>45	-	-	-	-	-	-	-	-
	Total	206			113				

Notes: Weight in kg and Stature in mm. All female subjects younger than 21 years old were measured.

*38 years-old is the highest age among females in the FAB. Delta = absolute difference in weight and stature.

Table 8. Summary statistics of anthropometric measurement for female FAB.

Dimensions	<i>n</i>	Mean	SE mean	SD	Min	Max	Skewness	Kurtosis	CV	CI
Abdominal ext depth, sitting	204	209.7	2.05	29.27	145	351	1.53	4.22	13.96	
*Acromial height, sitting	206	566.2	1.79	25.64	507	635	-0.01	0.06	4.53	±3.51
Biacromial breadth	206	351.9	1.31	18.84	295	408	-0.07	0.35	5.35	
Biceps circumference, flexed	204	280.4	1.95	27.86	213	420	1.36	3.78	9.94	
Bideltoid breadth	205	425.1	1.89	27.08	355	526	0.88	1.73	6.37	
Buttock circumference	205	992.5	5.29	75.77	828	1296	1.08	2.07	7.63	
*Buttock-knee length	206	582.9	1.83	26.24	520	685	0.45	0.93	4.50	±3.59
Buttock-popliteal length	206	475.5	1.5	21.5	421	542	0.4	0.47	4.52	
Calf circumference	205	364.6	1.83	26.21	295	452	0.45	0.65	7.19	
Cervical height	202	1400.3	3.57	50.70	1265	1532	-0.03	0.03	3.62	
Chest circumference	204	892.2	5.1	72.87	733	1155	0.94	1.70	8.17	
Chest depth	205	233.5	1.79	25.63	183	317	0.56	0.29	10.98	
Elbow rest height	204	230.7	1.54	22.03	164	295	-0.02	0.02	9.55	
*Eye height, sitting	206	751	2.05	29.13	678	815	-0.22	-0.29	3.88	±4.02
Foot breadth, horizontal	206	88.2	0.45	6.51	75	107	0.17	-0.59	7.38	
Foot length	205	247.1	0.83	11.97	218	278	0.18	-0.14	4.85	
Forearm circumference, flexed	205	248.8	1.18	16.85	204	312	0.74	1.32	6.77	
Forearm-hand length	205	435.8	1.33	19.05	392	492	0.11	-0.32	4.37	
Forearm-forearm breadth	203	432.2	2.96	42.18	340	615	0.99	2.51	9.76	
Functional leg length	200	1008	3.26	46.10	874	1149	-0.02	0.59	4.57	
Hand breadth	206	78.17	0.27	3.83	67	89	0	0.18	4.90	
Hand length	206	177.3	0.62	8.91	153	200	-0.12	-0.01	5.03	
Head circumference	206	549.4	1.01	14.55	511	590	0.11	-0.03	2.65	
Hip breadth, sitting	203	388.8	2.30	32.71	285	503	0.71	1.61	8.41	
*Knee height, sitting	206	513.7	1.45	20.78	455	568	0.13	-0.01	4.05	±2.84
Overhead fingertip reach, sitting	200	1286.9	3.37	47.70	1186	1410	-0.09	-0.56	3.70	
Palm length	206	102.7	0.36	5.21	89	117	0.16	-0.16	5.07	
Popliteal height	205	384.0	1.39	19.86	280	429	-0.95	3.45	5.17	
*Sitting height	206	865.8	2.19	31.07	787	949	-0.03	0	3.59	±4.28
Shoulder-elbow length	205	331.1	1.13	16.17	291	377	0.43	0.08	4.88	
Span	202	1649.1	4.76	67.6	1510	1865	0.39	0.03	4.10	
Stature	205	1642.3	3.79	54.30	1505	1784	-0.01	-0.18	3.31	±7.43
Thigh circumference	206	594.6	3.82	54.81	453	776	0.67	0.55	9.22	
Thigh clearance	204	168.9	1.13	16.19	126	214	0.12	-0.32	9.59	
Thigh max circumference, sitting	205	592.8	3.65	52.31	481	758	0.66	0.23	8.82	
*Thumbtip reach	206	709.5	2.54	36.21	605	803	-0.1	-0.33	5.10	±4.98
Waist circumference (narrower)	201	722.3	4.81	68.2	585	1008	1.22	2.74	9.44	
Waist circumference (Omphalion)	203	798	5.45	77.63	617	1116	1.18	2.72	9.73	
Weight (kg)	202	62.9	0.71	10.14	41	102	1.2	2.07	16.10	
BMI (kg/m ²)	202	23.3	0.24	3.33	17.6	37.5	1.46	3.01	14.30	

*cockpit critical dimensions. Weight in kg, BMI in kg/m² and, all other values in mm.

who could not be measured due to time conflicts and course assignments. Regarding geographical distribution of FAB survey participants compared to the FAB population, it was found very similar with most participants from Southeast Region (the most populous and developed in Brazil).

Additionally, despite Brazil's reputation as a country with extensive miscegenation, a substantial proportion of the sample declared themselves to be White (72%), while 22% identified themselves as Brown, and 4.4% as Black (Table 3). Note that race/ethnic comparison between the sample and the FAB population is not possible because there are no baseline population data available.

The sample included several types of aviation specialists (Fighter, Transportation, Patrol, Helicopter and others) in different military fields and branches, as illustrated in Table 4. Interestingly, the cadet population provided an important component. Unlike those training as pilots, cadets have not yet undergone rigid selection on the basis of their body size and proportions; therefore, this population had the potential to include more extreme body types.

The rank structure of the FAB survey participants is presented in Table 5. Age distribution is influenced by rank distribution, and as such, we also expect to find distinct age distributions within the ranks. The most frequently measured rank was Captain, closely followed by First Lieutenant, for both males and females. An important point to consider is that the highest rank achieved by the females who were measured was Major; consequently, the oldest female in this cohort was 38 years old.

Since it was not possible to measure the entire population as a census, the non-participants were examined regarding the reason for their absence as well as their weight and stature. Understanding both the reason for the non-participation as well as any variations in the anthropometric distribution of that cohort is also important. Tables 6 and 7 present the aforementioned information about the non-participants. Main reasons for non-participation were administrative – flight mission and missions abroad, course assignments; there are differences in the reasons between males and females. Furthermore, some military organisations/bases were not visited by the measurement

Table 9. Summary statistics of anthropometric measurements of male FAB.

Dimensions	<i>n</i>	Mean	SE mean	SD	Min	Max	Skewness	Kurtosis	CV	CI
Abdominal ext depth, sitting	2120	239.9	0.73	33.54	159	397	0.94	1.27	13.98	
*Acromial height, sitting	2122	598.4	0.65	29.78	509	699	0.10	-0.17	4.98	±1.27
Biacromial breadth	2122	394.8	0.47	21.60	313	465	-0.22	0.46	5.47	
Biceps circumference, flexed	2131	345.3	0.66	30.41	251	465	0.22	-0.02	8.81	
Bideltoid breadth	2121	483.6	0.61	28.20	365	596	0.13	0.66	5.83	
Buttock circumference	2116	989.4	1.45	66.48	770	1330	0.61	0.83	6.72	
*Buttock-knee length	2122	605.7	0.59	27.53	513	718	0.27	0.40	4.54	±1.17
Buttock-popliteal length	2121	490.1	0.55	25.41	400	602	0.20	0.50	5.18	
Calf circumference	2129	383.2	0.55	25.50	299	499	0.46	0.66	6.65	
Cervical height	2104	1504.4	1.28	58.50	1311	1697	0.04	0.06	3.89	
Chest depth	2121	249.1	0.49	22.95	175	335	0.26	0.41	9.21	
Chest circumference	2127	1014.7	1.55	71.70	806	1340	0.47	0.55	7.06	
Elbow rest height	2120	234.6	0.58	26.61	147	327	0.12	-0.13	11.34	
*Eye height, sitting	2122	802.1	0.68	31.15	688	917	0	0.07	3.88	±1.33
Foot breadth, horizontal	2123	96.9	0.14	6.65	77	120	0.10	-0.23	6.86	
Foot length	2122	270.8	0.29	13.58	219	319	0.06	0	5.01	
Forearm circumference, flexed	2132	298.7	0.44	20.28	241	399	0.37	0.86	6.79	
Forearm-hand length	2121	472.9	0.471	21.77	361	544	-0.07	0.46	4.60	
Forearm-forearm breadth	2118	516.3	0.96	44.23	390	646	0.02	-0.17	8.57	
Functional leg length	2074	1066.3	1.17	53.20	905	1242	0.13	0.02	4.99	
Hand breadth	2123	87.7	0.093	4.27	70	104	0.21	0.43	4.87	
Hand length	2123	190.4	0.23	10.35	157	237	0.10	0.09	5.44	
Head circumference	2132	568.7	0.34	15.57	509	644	0.18	0.72	2.74	
Hip breadth, sitting	2103	363.1	0.60	27.60	281	524	0.64	1.48	7.60	
*Knee height, sitting	2122	551.9	0.54	25.01	466	635	0.09	0.12	4.53	±1.06
Overhead fingertip reach, sitting	2074	1377.7	1.26	57.30	1210	1567	0.06	0.01	4.16	
Palm length	2123	112.1	0.14	6.38	88	134	0.07	0.17	5.69	
Popliteal height	2119	418.7	0.49	22.72	290	507	-0.01	0.77	5.43	
*Sitting height	2122	919.3	0.70	32.38	803	1024	-0.06	-0.09	3.52	±1.38
Shoulder-elbow length	2123	359.8	0.41	18.62	279	424	-0.11	0.32	5.17	
Span	2107	1791.4	1.63	74.90	1555	2050	0.09	-0.05	4.18	
Stature	2127	1755.2	1.34	62	1580	1953	0.03	-0.02	3.53	±2.64
Thigh circumference	2131	597.3	1.03	47.68	463	824	0.45	0.44	7.98	
Thigh clearance	2122	185.2	0.33	15.12	136	242	0.07	0.13	8.17	
Thigh max circumference, sitting	2131	596.3	0.99	45.93	453	790	0.51	0.51	7.70	
*Thumbtip reach	2122	779.6	0.88	40.38	650	922	0.07	0.10	5.18	±1.72
Waist circumference (narrower)	2127	852.1	1.82	84.09	674	1203	0.82	0.61	9.87	
Waist circumference (Omphalion)	2126	889.3	2.05	94.39	678	1259	0.75	0.49	10.61	
Weight (kg)	2116	79.2	0.25	11.67	47.6	133.9	0.78	1.14	14.74	
BMI (kg/m ²)	2111	25.6	0.07	3.25	15.5	41.6	0.82	1.58	12.65	

*cockpit critical dimensions. Weight in kg, BMI in kg/m² and, all other values in mm.

teams due to the restricted timeframe and the huge size of Brazil.

Data in Table 6 were gathered during data collection in each location, in coordination with the Personnel Management Sector. The non-participants' height and weight were collected from the FAB Annual Fitness Records. As can be seen, the weight and stature features between participants and non-participants are similar (see Table 7).

The summary anthropometric statistics of all 39 traditional measurements are shown in Tables 8–11 for males and females, separately. Tables 8 and 9 show the number of valid measurements (n), mean, standard error of the mean (SE Mean), standard deviation (SD), minimum and maximum measurements acquired, skewness, kurtosis and correlation variance (CV). The confidence intervals (CI) for a confidence level of 95% for the six cockpit design critical dimensions plus stature were also included in Tables 8 and 9. Tables 10 and 11 present the percentile distributions. Body Mass Index (BMI) is also presented in the aforementioned tables.

4. Discussion

It is noteworthy to clarify why this study still used manually measured body-size variables obtained by using traditional instruments. Firstly, because these methods, when properly employed and controlled, still are a highly accurate means of documenting the body sizes of individuals and of whole populations. Recent large-scale surveys of military populations, such as the 2010 Anthropometric Survey of US Marine Corps Personnel (Gordon et al. 2013) and the 2012 Anthropometric Survey of US Army Personnel (Gordon et al. 2014), still include traditional anthropometric measurements in order to facilitate population comparisons over time and between user groups. In addition, methods for using 3D imaging systems are not standardised and scan-extracted circumference measurements are much more difficult to obtain reliably.

In accordance with previous research (Gordon et al. 2013; ISO 2012) the sample can be considered truly representative of the overall FAB pilot population as well as large enough to accurately reflect the variability in this

Table 10. Anthropometric percentile summary statistics, female.

Dimensions	1%	2%	5%	10%	25%	50%	75%	90%	95%	98%	99%
Abdominal ext depth, sitting	161	163	173	179	191	205	222	243	270	301	326
*Acromial height, sitting	508	509	520	530	551	566	584	595	610	624	631
Biacromial breadth	305	307	319	330	339	351	365	375	382	393	404
Biceps circumference, flexed	231	234	245	252	261	276	292	317	335	362	380
Bideltoid breadth	357	372	390	39	407	422	437	461	482	497	522
Buttock circumference	844	852	891	914	945	981	1028	1101	1137	1205	1272
*Buttock-knee length	520	526	544	551	566	581	598	618	630	643	658
Buttock-popliteal length	423	435	442	449	461	474	488	504	514	530	540
Calf circumference	304	312	326	334	347	362	382	401	409	424	449
Cervical height	1276	1290	1314	1334	1368	1400	1430	1462	1484	1511	1529
Chest circumference	741	771	792	809	841	886	932	975	1041	1118	1146
Chest depth	186	190	195	203	214	232	250	264	283	300	305
Elbow rest height	176	186	194	201	217	231	246	259	268	276	280
*Eye height, sitting	678	683	698	713	732	754	773	787	799	811	814
Foot breadth, horizontal	75	76	78	80	83	88	94	97	99	102	103
Foot length	220	223	229	232	239	247	255	262	269	274	278
Forearm circumference, flexed	206	222	226	231	237	247	258	270	282	292	307
Forearm-hand length	394	396	406	411	422	435	450	462	467	475	480
Forearm-forearm breadth	351	355	372	382	405	427	454	484	505	551	601
Functional leg length	876	900	931	953	981	1011	1034	1065	1087	1103	1128
Hand breadth	69	70	72	73	76	78	81	83	84	87	88
Hand length	154	156	161	166	171	177	184	189	192	195	198
Head circumference	516	520	525	530	540	548	559	569	575	582	588
Hip breadth, sitting	315	325	346	353	367	387	405	430	450	487	490
*Knee height, sitting	467	472	478	488	499	513	528	540	551	563	567
Overhead fingertip reach, sit.	1187	1192	1207	1218	1253	1290	1320	1347	1365	1378	1387
Palm length	90	92	94	96	99	102	106	110	112	113	116
Popliteal height	314	344	349	358	373	386	396	408	415	420	423
*Sitting height	787	799	813	822	846	869	886	906	917	936	940
Shoulder-elbow length	300	302	305	312	319	330	341	353	358	373	378
Span	1511	1530	1544	1560	1603	1645	1700	1738	1768	1804	1839
Stature	1516	1533	1547	1569	1605	1644	1679	1708	1729	1760	1778
Thigh circumference	493	509	521	532	554	588	624	671	694	732	763
Thigh clearance	130	135	144	150	156	168	180	193	195	202	205
Thigh max circumference, sitting	490	507	521	532	557	586	620	674	696	720	738
*Thumbtip reach	630	631	650	660	685	710	737	754	770	785	787
Waist circumference (narrower)	595	613	626	653	679	712	755	809	859	906	998
Waist circumference (Omphalion)	659	662	696	716	746	787	835	892	941	1024	1108
Weight (kg)	45.1	47.1	50.4	52.7	55.9	61.3	67.6	77.5	83.9	94.2	102.1
BMI (kg/m ²)	18.1	18.8	19.4	19.8	21.0	22.6	24.7	28.1	30.0	34.9	37.4

*cockpit critical dimensions. Weight in kg, BMI in kg/m² and, all other values in mm.

population. Power analysis using equations 1 and 2 from ISO 15535 (2012) showed that male sample size was well above the minimum required and the female sample size was above for the cockpit critical dimensions. The presentation of representative female pilot data is relevant for cockpit design and several military equipment and uniforms, since most previous studies have been presented only anthropometric data from male subjects.

The sample demographic characteristics were very similar to the entire Brazilian Air Force population. In the same way, the comparison between participants and non-participants revealed that weight and stature of both groups were similar.

This study captured a broad age range. A common sample bias in many military research studies occurs with the age distribution, as younger service members are more likely to be available to participate in studies than senior individuals. This has the potential to create problems, since younger military personnel tend to be more fit and thinner, thus skewing the sample data. In our study we captured a large proportion of higher ranking officials (Lieutenant

Colonel and Colonel), measuring a total of 341 subjects (14.6%) above 40 years old.

One explanation for this may have been due to the fact that the FAB Survey was an official event, established by an Ordinance that described the importance of the Survey, and encouraging all pilots to participate.

The sample was also geographically representative. Only 2 out of 26 states (Acre and Tocantins) had no representatives in the sample population. The most common birthplace for both male and female participants was Rio de Janeiro.

The age distribution of FAB survey participants was very similar to the whole FAB population distribution. Since there have been no studies to date describing the race/ethnicity distribution in the FAB, nor is it registered in the databases, it was not possible to determine whether the race/ethnicity distribution in the sample is representative of the actual ethnic distribution. This survey creates a starting point from which the FAB may choose to initiate further investigations.

Table 11. Anthropometric percentile summary statistics, male.

Dimensions	1%	2%	5%	10%	25%	50%	75%	90%	95%	98%	99%
Abdominal ext depth, sitting	178	186	196	203	217	234	259	284	305	328	343
*Acromial height, sitting	533	540	550	560	578	598	619	637	649	660	669
Biacromial breadth	337	345		368	381	395	410	421	429	437	446
Biceps circumference, flexed	280	285	297	307	323	344	366	385	395	411	422
Bideltoid breadth	415	427	441	450	465	482	502	520	530	544	555
Buttock circumference	859	874	893	911	943	984	1030	1077	1106	1154	1171
*Buttock-knee length	546	552	563	572	587	605	623	640	653	668	680
Buttock-popliteal length	432	439	450	459	473	490	506	522	532	547	556
Calf circumference	330	36	345	353	366	381	398	416	428	442	450
Cervical height	1371	1383	1404	1428	1466	1505	1542	1578	1601	1627	1650
Chest circumference	868	881	908	930	964	1011	1059	1104	1140	1187	1214
Chest depth	196	202	214	221	234	249	262	279	288	301	311
Elbow rest height	178	182	191	202	215	234	252	270	281	291	297
*Eye height, sitting	730	739	750	762	781	802	823	841	853	864	878
Foot breadth, horizontal	82	84	86	88	92	97	101	106	108	110	112
Foot length	240	243	250	254	261	270	280	288	294	299	303
Forearm circumference, flexed	254	259	267	275	285	298	312	324	332	343	350
Forearm-hand length	422	429	437	446	459	473	487	500	510	520	525
Forearm-forearm breadth	413	428	443	459	486	516	547	573	590	609	622
Functional leg length	943	963	979	998	1029	1066	1100	1136	1159	1182	1194
Hand breadth	79	80	81	82	85	87	90	93	95	97	98
Hand length	168	170	173	177	183	190	197	204	208	212	215
Head circumference	533	537	545	550	559	569	578	589	595	603	608
Hip breadth, sitting	305	313	322	330	345	361	379	398	413	426	440
*Knee height, sitting	495	502	511	519	535	552	568	584	594	607	614
Overhead fingertip reach, sit.	1250	1263	1282	1303	1338	1379	1415	1451	1474	1501	1516
Palm length	97	100	102	104	108	112	116	120	123	125	128
Popliteal height	364	373	383	390	404	419	433	447	456	469	476
*Sitting height	843	853	865	877	898	920	942	961	972	984	993
Shoulder-elbow length	313	320	329	335	349	360	372	383	390	397	405
Span	1628	1640	1668	1693	1740	1790	1841	1889	1915	1950	1971
Stature	1613	1626	1651	1674	1713	1755	1797	1833	1857	1886	1908
Thigh circumference	503	510	525	540	564	594	628	660	685	701	719
Thigh clearance	149	153	161	166	175	185	195	205	210	216	221
Thigh max circumference, sitting	505	514	528	540	564	593	624	656	680	703	720
*Thumbtip reach	690	700	715	730	751	780	806	830	848	868	878
Waist circumference (narrower)	707	724	740	757	790	840	899	970	1018	1066	1097
Waist circumference (Omphalion)	729	741	761	781	819	879	944	101	1070	1128	1154
Weight (kg)	58.6	60.2	62.7	65.7	71.1	77.9	85.8	94.8	100.2	107.8	113.1
BMI (kg/m ²)	19.6	20.1	21.0	21.9	23.4	25.3	27.4	29.8	31.5	33.8	35.6

*cockpit critical dimensions. Weight in kg, BMI in kg/m² and, all other values in mm.

When controlling non-participants regarding stature and weight there was no significant difference between these groups.

Another important contribution of this study is the data provided beyond the 5th/95th percentile anthropometric values, since just a few studies presents this relevant information (Parkinson and Garneau 2016), as well as the Skewness, Kurtosis and CV values necessary for data estimation and comparison (Roebuck 1995).

The newly created FAB database will allow engineers and designers to develop more precise protective equipment, workplaces and uniforms. The database also serves as a comprehensive and reliable resource for anthropometric research, since most available resources that engineers, designers and practioners have been exposed to them, only shows inaccurate, outdated and/or incomplete data, as mentioned by Parkinson and Garneau (2016).

The results of this study have numerous implications for the FAB, in particular concerning the development and/or acquisition of new aircraft, upgrading existing aircraft, and the design and sizing of uniforms and protective equipment worn by aircrew. This study provides critical data to address cockpit design, space arrangement issues and adjustments. The data will be also useful for protective gear and equipment design, as well as for digital human modelling often used in early cockpit design processes.

Another important implication is that the present study can be considered as a starting point for the improvement of the accommodation of females in cockpit design in Brazil, reducing gender bias and the impact in women's career as pilots.

5. Conclusion

This research provided the first reported anthropometric information for Brazilian Air Force pilots. The study investigated 39 anthropometric body dimensions that are of relevance to aircraft cockpit design for ensuring ergonomic safety and comprehensive pilot accommodation, effective protective gear and uniform design, as well as more accurate digital human modelling.

The survey recruited a total of 2339 participants, a large sample that included 70% of the entire FAB pilot population, which was representative for the point of size, geographic and age distribution, as well as has not presented differences from the non-participants.

Finally, the results of this study have numerous implications for the FAB, especially implications for human-equipment fit. The up-to-date pilots' anthropometric data and multivariate anthropometric models will benefit the design of future aircraft cockpits which, in turn, will help promote the safety, comfort and performance of all pilots, facilitating the accomplishment of their mission.

Being the first large-scale anthropometric survey in the Brazilian Armed Forces, the data obtained in this study provide more accurate anthropometric information about the FAB pilots.

Therefore, this study contributes to general human engineering design criteria for military systems, equipment and facilities, thus ensuring optimum accommodation of the FAB personnel. Including these data in the anthropometric criteria for a new FAB aircraft would lead to a more inclusive air force, minimising gender bias.

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